

Which orange is grasped vs. which was requested

Rows: requested position. Columns: position grasped. Cells: % of the row's grasps. Scene states 0-1 pooled.

Standard fine-tuning

action expert + state projection

67% obeyed n = 195 grasps					
	Left	Middle	Right	Top right	Bottom right
Left	97 n=33	3			
Middle	15	80 n=46	4		
Right		52	41 n=71	6	1
Top right	4			96 n=25	
Bottom right		5	5	42	47 n=19

LM-tuned

+ language model

67% obeyed n = 206 grasps					
	Left	Middle	Right	Top right	Bottom right
Left	92 n=24	8			
Middle	11	89 n=63			
Right	2	64	32 n=56	2	
Top right	3			89 n=38	8
Bottom right		4	8	62	25 n=24

Fully-tuned

+ language model + vision encoder

64% obeyed n = 200 grasps					
	Left	Middle	Right	Top right	Bottom right
Left	100 n=35				
Middle	13	78 n=55	9		
Right	2	66	27 n=56	4	2
Top right	6			84 n=32	9
Bottom right	11		11	44	33 n=18

73% obeyed n = 203 grasps					
	Left	Middle	Right	Top right	Bottom right
Left	97 n=30	3			
Middle	16	79 n=62	2	3	
Right		48	47 n=58	3	2
Top right				96 n=28	4
Bottom right		5		29	67 n=21

67% obeyed n = 181 grasps					
	Left	Middle	Right	Top right	Bottom right
Left	71 n=42	17		12	
Middle	17	75 n=52	4	4	
Right	10	36	46 n=39	5	3
Top right				93 n=27	7
Bottom right				47	53 n=17

 **Obeyed**
grasped the requested orange

 **Misgrab**
grasped a different orange

Cell value: % of the row's grasps;
darker shade = larger share.

No fully-tuned run exists for
Teleop+Auto.

Teleop

Teleop+Auto